

Chapter 7 Worked Examples Study Guide

Eigenvalues and Eigenvectors

Based on Ron Larson, *Elementary Linear Algebra*, 8th Edition, Chapter 7

Compiled for step-by-step exam preparation

Covers: 7.1, 7.2, 7.3, and 7.4

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How to Use This PDF

This PDF gathers every worked example from the Chapter 7 tutoring sequence. Each solution follows the textbook method: find eigenvalues using the characteristic equation, find eigenvectors using homogeneous systems, then apply diagonalization, orthogonal diagonalization, and applications.

Core formulas:

$$Ax = \lambda x, \quad \det(\lambda I - A) = 0, \quad (\lambda I - A)x = 0.$$

For diagonalization:

$$P^{-1}AP = D.$$

For orthogonal diagonalization:

$$P^T AP = D.$$

1 7.1 Eigenvalues and Eigenvectors

1.1 Example 7.1.1: Verifying Eigenvalues and Eigenvectors

Given

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}, \quad x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Verify the corresponding eigenvalues.

Solution for x_1 :

$$Ax_1 = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2(1) + 0(0) \\ 0(1) + (-1)(0) \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Factor out 2:

$$\begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2x_1.$$

Therefore,

$$Ax_1 = 2x_1,$$

so x_1 is an eigenvector corresponding to $\lambda = 2$.

Solution for x_2 :

$$Ax_2 = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2(0) + 0(1) \\ 0(0) + (-1)(1) \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Factor out -1 :

$$\begin{bmatrix} 0 \\ -1 \end{bmatrix} = -1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -x_2.$$

Therefore,

$$Ax_2 = -1x_2,$$

so x_2 is an eigenvector corresponding to $\lambda = -1$.

1.2 Example 7.1.2: Verifying Eigenvectors and Finding Eigenvalues

Given

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad x_1 = \begin{bmatrix} -3 \\ -1 \\ 1 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Verify that x_1 and x_2 are eigenvectors and find their eigenvalues.

For x_1 :

$$Ax_1 = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ -1 \\ 1 \end{bmatrix}.$$

Compute each entry:

$$\text{Row 1 : } 1(-3) + (-2)(-1) + 1(1) = -3 + 2 + 1 = 0,$$

$$\text{Row 2 : } 0(-3) + 0(-1) + 0(1) = 0,$$

$$\text{Row 3 : } 0(-3) + 1(-1) + 1(1) = -1 + 1 = 0.$$

So

$$Ax_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0x_1.$$

Therefore, x_1 is an eigenvector corresponding to

$$\lambda = 0.$$

For x_2 :

$$Ax_2 = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 1x_2.$$

Therefore, x_2 is an eigenvector corresponding to

$$\lambda = 1.$$

1.3 Example 7.1.3: Finding Eigenspaces Geometrically

Given

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Find the eigenvalues and eigenspaces.

Let

$$v = \begin{bmatrix} x \\ y \end{bmatrix}.$$

Then

$$Av = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}.$$

This reflects a vector across the y -axis.

Vectors on the x -axis:

$$v = \begin{bmatrix} x \\ 0 \end{bmatrix}.$$

Then

$$Av = \begin{bmatrix} -x \\ 0 \end{bmatrix} = -1 \begin{bmatrix} x \\ 0 \end{bmatrix}.$$

So the eigenvalue is

$$\lambda = -1,$$

and the eigenspace is

$$E_{-1} = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}.$$

Vectors on the y -axis:

$$v = \begin{bmatrix} 0 \\ y \end{bmatrix}.$$

Then

$$Av = \begin{bmatrix} 0 \\ y \end{bmatrix} = 1 \begin{bmatrix} 0 \\ y \end{bmatrix}.$$

So the eigenvalue is

$$\lambda = 1,$$

and the eigenspace is

$$E_1 = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}.$$

1.4 Example 7.1.4: Finding Eigenvalues and Eigenvectors of a 2×2 Matrix

Given

$$A = \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix}.$$

Find the eigenvalues and eigenvectors.

Step 1: Form $\lambda I - A$.

$$\lambda I - A = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 2 & -12 \\ 1 & -5 \end{bmatrix} = \begin{bmatrix} \lambda - 2 & 12 \\ -1 & \lambda + 5 \end{bmatrix}.$$

Step 2: Find the characteristic equation.

$$\begin{aligned} \det(\lambda I - A) &= \begin{vmatrix} \lambda - 2 & 12 \\ -1 & \lambda + 5 \end{vmatrix} = (\lambda - 2)(\lambda + 5) - 12(-1). \\ &= (\lambda - 2)(\lambda + 5) + 12 = \lambda^2 + 3\lambda - 10 + 12 = \lambda^2 + 3\lambda + 2. \end{aligned}$$

Factor:

$$\lambda^2 + 3\lambda + 2 = (\lambda + 1)(\lambda + 2).$$

Thus

$$(\lambda + 1)(\lambda + 2) = 0,$$

so

$$\lambda = -1, \quad \lambda = -2.$$

Step 3: Eigenvectors for $\lambda = -1$.

$$(-1)I - A = \begin{bmatrix} -3 & 12 \\ -1 & 4 \end{bmatrix}.$$

Solve

$$\begin{bmatrix} -3 & 12 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The equation is

$$-x_1 + 4x_2 = 0,$$

so

$$x_1 = 4x_2.$$

Let $x_2 = t$. Then

$$x = \begin{bmatrix} 4t \\ t \end{bmatrix} = t \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad t \neq 0.$$

Therefore,

$$E_{-1} = \text{span} \left\{ \begin{bmatrix} 4 \\ 1 \end{bmatrix} \right\}.$$

Step 4: Eigenvectors for $\lambda = -2$.

$$(-2)I - A = \begin{bmatrix} -4 & 12 \\ -1 & 3 \end{bmatrix}.$$

Solve

$$\begin{bmatrix} -4 & 12 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The equation is

$$-x_1 + 3x_2 = 0,$$

so

$$x_1 = 3x_2.$$

Let $x_2 = t$. Then

$$x = \begin{bmatrix} 3t \\ t \end{bmatrix} = t \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad t \neq 0.$$

Therefore,

$$E_{-2} = \text{span} \left\{ \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right\}.$$

1.5 Example 7.1.5: Repeated Eigenvalue and Eigenspace Dimension

Given

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Find the eigenvalues and eigenspace dimension.

Step 1: Characteristic equation.

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & -1 & 0 \\ 0 & \lambda - 2 & 0 \\ 0 & 0 & \lambda - 2 \end{bmatrix}.$$

Since the matrix is triangular,

$$\det(\lambda I - A) = (\lambda - 2)^3.$$

Thus

$$\lambda = 2$$

with algebraic multiplicity 3.

Step 2: Find the eigenspace.

$$2I - A = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Solve

$$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

The only equation is

$$-x_2 = 0,$$

so

$$x_2 = 0.$$

Let

$$x_1 = s, \quad x_3 = t.$$

Then

$$x = \begin{bmatrix} s \\ 0 \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

A basis for the eigenspace is

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Therefore,

$$\dim(E_2) = 2.$$

Important idea:

eigenspace dimension $2 <$ algebraic multiplicity 3.

1.6 Example 7.1.6: Eigenvalues and Bases for Eigenspaces of a 4×4 Matrix

Given

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -10 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}.$$

The characteristic polynomial is

$$\det(\lambda I - A) = (\lambda - 1)^2(\lambda - 2)(\lambda - 3).$$

Thus the eigenvalues are

$$\lambda = 1, \quad \lambda = 2, \quad \lambda = 3.$$

For $\lambda = 1$: Solving $(I - A)x = 0$ gives

$$x_1 + 2x_4 = 0, \quad x_3 - 2x_4 = 0.$$

Therefore,

$$x_1 = -2x_4, \quad x_3 = 2x_4.$$

Let

$$x_2 = s, \quad x_4 = t.$$

Then

$$x = \begin{bmatrix} -2t \\ s \\ 2t \\ t \end{bmatrix} = s \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \end{bmatrix}.$$

So

$$E_1 = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \end{bmatrix} \right\}.$$

For $\lambda = 2$: Solving $(2I - A)x = 0$ gives

$$x_1 = 0, \quad x_2 - 5x_3 = 0, \quad x_4 = 0.$$

So

$$x_2 = 5x_3.$$

Let $x_3 = t$. Then

$$x = \begin{bmatrix} 0 \\ 5t \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} 0 \\ 5 \\ 1 \\ 0 \end{bmatrix}.$$

Thus

$$E_2 = \text{span} \left\{ \begin{bmatrix} 0 \\ 5 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

For $\lambda = 3$: Solving $(3I - A)x = 0$ gives

$$x_1 = 0, \quad x_2 + 5x_4 = 0, \quad x_3 = 0.$$

So

$$x_2 = -5x_4.$$

Let $x_4 = t$. Then

$$x = \begin{bmatrix} 0 \\ -5t \\ 0 \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ -5 \\ 0 \\ 1 \end{bmatrix}.$$

Thus

$$E_3 = \text{span} \left\{ \begin{bmatrix} 0 \\ -5 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

1.7 Example 7.1.7: Eigenvalues of Triangular and Diagonal Matrices

Part A. Given

$$A = \begin{bmatrix} 2 & -1 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & -3 \end{bmatrix}.$$

Because A is triangular, its eigenvalues are the diagonal entries:

$$\lambda = 2, \quad 1, \quad -3.$$

To confirm:

$$\lambda I - A = \begin{bmatrix} \lambda - 2 & 1 & -5 \\ 0 & \lambda - 1 & -3 \\ 0 & 0 & \lambda + 3 \end{bmatrix}.$$

Since this matrix is triangular,

$$\det(\lambda I - A) = (\lambda - 2)(\lambda - 1)(\lambda + 3).$$

So

$$\lambda = 2, \quad 1, \quad -3.$$

Part B. If a diagonal matrix has diagonal entries

$$-1, \quad 2, \quad 0, \quad -4, \quad 3,$$

then its eigenvalues are exactly those entries:

$$\lambda = -1, \quad 2, \quad 0, \quad -4, \quad 3.$$

1.8 Example 7.1.8: Eigenvalues and Eigenspaces of a Linear Transformation Matrix

Given

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

Find eigenvalues and bases for eigenspaces.

Step 1: Characteristic equation.

$$\lambda I - A = \begin{bmatrix} \lambda - 1 & -3 & 0 \\ -3 & \lambda - 1 & 0 \\ 0 & 0 & \lambda + 2 \end{bmatrix}.$$

Then

$$\begin{aligned} \det(\lambda I - A) &= \begin{vmatrix} \lambda - 1 & -3 \\ -3 & \lambda - 1 \end{vmatrix} (\lambda + 2). \\ &= [(\lambda - 1)^2 - 9](\lambda + 2). \end{aligned}$$

Use difference of squares:

$$(\lambda - 1)^2 - 9 = (\lambda - 4)(\lambda + 2).$$

So

$$\det(\lambda I - A) = (\lambda - 4)(\lambda + 2)^2.$$

Thus

$$\lambda = 4, \quad \lambda = -2.$$

For $\lambda = 4$:

$$4I - A = \begin{bmatrix} 3 & -3 & 0 \\ -3 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}.$$

The equations give

$$x_1 = x_2, \quad x_3 = 0.$$

Let $x_2 = t$. Then

$$x = t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

So

$$E_4 = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

For $\lambda = -2$:

$$-2I - A = \begin{bmatrix} -3 & -3 & 0 \\ -3 & -3 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The equation is

$$x_1 + x_2 = 0,$$

so $x_1 = -x_2$, and x_3 is free. Let $x_2 = s$ and $x_3 = t$. Then

$$x = \begin{bmatrix} -s \\ s \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

A convenient basis is

$$E_{-2} = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

2 7.2 Diagonalization

2.1 Example 7.2.1: Verifying a Diagonalization

Given

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Show that $P^{-1}AP$ is diagonal.

Step 1: Find P^{-1} . The top-left block

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

has determinant

$$1(-1) - 1(1) = -2.$$

Its inverse is

$$\frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}.$$

Therefore,

$$P^{-1} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Step 2: Compute AP .

$$AP = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Multiplying column by column:

$$A \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix}, \quad A \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \\ 0 \end{bmatrix}, \quad A \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}.$$

Thus

$$AP = \begin{bmatrix} 4 & -2 & 0 \\ 4 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

Step 3: Compute $P^{-1}AP$.

$$P^{-1}AP = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & -2 & 0 \\ 4 & 2 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

Compute entries:

$$P^{-1}AP = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

Therefore, A is diagonalizable and

$$D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

2.2 Example 7.2.2: Eigenvalues of Similar Matrices

Suppose

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 1 \\ -1 & -2 & 4 \end{bmatrix}$$

is similar to

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Because similar matrices have the same eigenvalues, and D is diagonal, the eigenvalues of A are

$$\lambda = 1, \quad 2, \quad 3.$$

Check by characteristic polynomial:

$$\lambda I - A = \begin{bmatrix} \lambda - 1 & 0 & 0 \\ 1 & \lambda - 1 & -1 \\ 1 & 2 & \lambda - 4 \end{bmatrix}.$$

Expand along the first row:

$$\begin{aligned} \det(\lambda I - A) &= (\lambda - 1) \begin{vmatrix} \lambda - 1 & -1 \\ 2 & \lambda - 4 \end{vmatrix} \\ &= (\lambda - 1)[(\lambda - 1)(\lambda - 4) - (-1)(2)] \\ &= (\lambda - 1)[(\lambda - 1)(\lambda - 4) + 2]. \end{aligned}$$

Expand:

$$(\lambda - 1)(\lambda - 4) + 2 = \lambda^2 - 5\lambda + 4 + 2 = \lambda^2 - 5\lambda + 6.$$

Factor:

$$\lambda^2 - 5\lambda + 6 = (\lambda - 2)(\lambda - 3).$$

So

$$\det(\lambda I - A) = (\lambda - 1)(\lambda - 2)(\lambda - 3),$$

and the eigenvalues are 1, 2, 3.

2.3 Example 7.2.3: Diagonalizable Matrix from Known Eigenvectors

For

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix},$$

from Section 7.1 the eigenvectors are

$$p_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad (\lambda = 4), \quad p_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad (\lambda = -2), \quad p_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (\lambda = -2).$$

Form

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The matching diagonal matrix is

$$D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}.$$

Therefore,

$$P^{-1}AP = D.$$

The important rule is that the order of columns in P must match the order of eigenvalues in D .

2.4 Example 7.2.4: Diagonalizing a 3×3 Matrix

Show that

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 1 \\ -3 & 1 & -1 \end{bmatrix}$$

is diagonalizable and find P such that $P^{-1}AP$ is diagonal.

The characteristic polynomial is

$$\det(\lambda I - A) = (\lambda - 2)(\lambda + 2)(\lambda - 3).$$

So the eigenvalues are

$$\lambda = 2, \quad -2, \quad 3.$$

Since the eigenvalues are distinct, the matrix is diagonalizable. We still find eigenvectors.

For $\lambda = 2$:

$$2I - A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -1 & -1 \\ 3 & -1 & 3 \end{bmatrix}.$$

Row-reducing gives

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$x_1 + x_3 = 0, \quad x_2 = 0.$$

Let $x_3 = t$. Then

$$x = t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Choose

$$p_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

For $\lambda = -2$:

$$-2I - A = \begin{bmatrix} -3 & 1 & 1 \\ -1 & -5 & -1 \\ 3 & -1 & -1 \end{bmatrix}.$$

Row-reducing gives

$$\begin{bmatrix} 1 & 0 & -1/4 \\ 0 & 1 & 1/4 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$x_1 = \frac{1}{4}x_3, \quad x_2 = -\frac{1}{4}x_3.$$

Let $x_3 = 4t$. Then

$$x = t \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}.$$

Choose

$$p_2 = \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}.$$

For $\lambda = 3$:

$$3I - A = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 0 & -1 \\ 3 & -1 & 4 \end{bmatrix}.$$

Row-reducing gives

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Thus

$$x_1 = -x_3, \quad x_2 = x_3.$$

Let $x_3 = t$. Then

$$x = t \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$$

Choose

$$p_3 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$$

Now form

$$P = \begin{bmatrix} -1 & 1 & -1 \\ 0 & -1 & 1 \\ 1 & 4 & 1 \end{bmatrix}.$$

The matching diagonal matrix is

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Thus

$$P^{-1}AP = D.$$

2.5 Example 7.2.5: Diagonalizing a 4×4 Matrix

Given

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -10 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}.$$

From Section 7.1, eigenvectors are

$$\lambda = 1: \quad \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} -2 \\ 0 \\ 2 \\ 1 \end{bmatrix},$$

$$\lambda = 2: \quad \begin{bmatrix} 0 \\ 5 \\ 1 \\ 0 \end{bmatrix},$$

$$\lambda = 3: \quad \begin{bmatrix} 0 \\ -5 \\ 0 \\ 1 \end{bmatrix}.$$

There are four linearly independent eigenvectors for a 4×4 matrix, so A is diagonalizable.

Form

$$P = \begin{bmatrix} 0 & -2 & 0 & 0 \\ 1 & 0 & 5 & -5 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

The first two columns correspond to $\lambda = 1$, the third to $\lambda = 2$, and the fourth to $\lambda = 3$. Therefore

$$D = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

Thus

$$P^{-1}AP = D.$$

2.6 Example 7.2.6: A Matrix That Is Not Diagonalizable

Show that

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

is not diagonalizable.

Because A is triangular, its eigenvalue is

$$\lambda = 1$$

with multiplicity 2.

Find the eigenspace:

$$I - A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 0 & 0 \end{bmatrix}.$$

Solve

$$\begin{bmatrix} 0 & -2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The equation is

$$-2x_2 = 0,$$

so

$$x_2 = 0.$$

Let $x_1 = t$. Then

$$x = t \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

There is only one linearly independent eigenvector, but a 2×2 matrix needs two. Therefore,

A is not diagonalizable.

2.7 Example 7.2.7: Using the Distinct Eigenvalue Shortcut

Given

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -3 \end{bmatrix}.$$

Because A is triangular, the eigenvalues are the diagonal entries:

$$\lambda = 1, \quad 0, \quad -3.$$

They are three distinct eigenvalues for a 3×3 matrix. Therefore,

A is diagonalizable.

No eigenvector calculation is required because n distinct eigenvalues guarantee diagonalizability.

2.8 Example 7.2.8: Diagonal Basis for a Linear Transformation

Let

$$T(x_1, x_2, x_3) = (x_1 - x_2 - x_3, x_1 + 3x_2 + x_3, -3x_1 + x_2 - x_3).$$

Find a basis B for $\mathbb{R}^3$ such that the matrix of T relative to B is diagonal.

The standard matrix is

$$A = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 3 & 1 \\ -3 & 1 & -1 \end{bmatrix}.$$

From Example 7.2.4, eigenvectors are

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$$

So a basis is

$$B = \{(-1, 0, 1), (1, -1, 4), (-1, 1, 1)\}.$$

The matrix of T relative to this basis is

$$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

3 7.3 Symmetric Matrices and Orthogonal Diagonalization

3.1 Example 7.3.1: Checking Symmetric Matrices

A matrix A is symmetric when

$$A = A^T.$$

Matrix 1:

$$A = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 3 & 0 \\ -2 & 0 & 5 \end{bmatrix}.$$

Transpose:

$$A^T = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 3 & 0 \\ -2 & 0 & 5 \end{bmatrix} = A.$$

So A is symmetric.

Matrix 2:

$$B = \begin{bmatrix} 4 & 3 \\ 3 & 1 \end{bmatrix}.$$

Transpose:

$$B^T = \begin{bmatrix} 4 & 3 \\ 3 & 1 \end{bmatrix} = B.$$

So B is symmetric.

Matrix 3:

$$C = \begin{bmatrix} 3 & 1 & 1 \\ 2 & -4 & 0 \\ 1 & 0 & 5 \end{bmatrix}.$$

Here

$$c_{12} = 1, \quad c_{21} = 2.$$

Since $c_{12} \neq c_{21}$, C is not symmetric.

3.2 Example 7.3.2: A Nonsymmetric Matrix with Imaginary Eigenvalues

Consider

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Find its eigenvalues.

$$\lambda I - A = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} \lambda & -1 \\ 1 & \lambda \end{bmatrix}.$$

Then

$$\det(\lambda I - A) = \lambda^2 - (-1)(1) = \lambda^2 + 1.$$

Set equal to zero:

$$\lambda^2 + 1 = 0.$$

Thus

$$\lambda^2 = -1,$$

so

$$\lambda = \pm i.$$

This shows that nonsymmetric matrices may have non-real eigenvalues.

3.3 Example 7.3.3: Every 2×2 Symmetric Matrix Is Diagonalizable

Let

$$A = \begin{bmatrix} a & c \\ c & b \end{bmatrix}.$$

This is symmetric.

Step 1: Form the characteristic polynomial.

$$\lambda I - A = \begin{bmatrix} \lambda - a & -c \\ -c & \lambda - b \end{bmatrix}.$$

$$\det(\lambda I - A) = (\lambda - a)(\lambda - b) - c^2.$$

Expand:

$$(\lambda - a)(\lambda - b) - c^2 = \lambda^2 - (a + b)\lambda + ab - c^2.$$

Step 2: Study the discriminant. The discriminant is

$$[-(a + b)]^2 - 4(1)(ab - c^2).$$

Simplify:

$$(a + b)^2 - 4ab + 4c^2 = a^2 - 2ab + b^2 + 4c^2 = (a - b)^2 + 4c^2.$$

This is always nonnegative.

If the discriminant is positive, the matrix has two distinct real eigenvalues, so it is diagonalizable.

If the discriminant is zero, then

$$(a - b)^2 + 4c^2 = 0$$

forces

$$a = b, \quad c = 0.$$

Then

$$A = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix},$$

which is already diagonal. Therefore every 2×2 symmetric matrix is diagonalizable.

3.4 Example 7.3.4: Eigenspace Dimensions of a Symmetric Matrix

Given

$$A = \begin{bmatrix} 1 & -2 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -2 & 1 \end{bmatrix}.$$

The characteristic polynomial is

$$\det(\lambda I - A) = (\lambda + 1)^2(\lambda - 3)^2.$$

So the eigenvalues are

$$\lambda = -1, \quad \lambda = 3,$$

each with multiplicity 2. Because A is symmetric, each eigenspace has dimension 2.

For $\lambda = -1$: Solve $(A + I)x = 0$:

$$A + I = \begin{bmatrix} 2 & -2 & 0 & 0 \\ -2 & 2 & 0 & 0 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & -2 & 2 \end{bmatrix}.$$

The equations imply

$$x_1 = x_2, \quad x_3 = x_4.$$

Let $x_2 = s$ and $x_4 = t$. Then

$$x = \begin{bmatrix} s \\ s \\ t \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}.$$

Thus

$$E_{-1} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

So $\dim(E_{-1}) = 2$.

For $\lambda = 3$:

$$3I - A = \begin{bmatrix} 2 & 2 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 2 & 2 \end{bmatrix}.$$

The equations imply

$$x_1 + x_2 = 0, \quad x_3 + x_4 = 0.$$

So

$$x_1 = -x_2, \quad x_3 = -x_4.$$

Let $x_2 = s$ and $x_4 = t$. Then

$$x = \begin{bmatrix} -s \\ s \\ -t \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

A convenient basis is

$$E_3 = \text{span} \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

So $\dim(E_3) = 2$.

3.5 Example 7.3.5: Checking an Orthogonal Matrix

Given

$$P = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Show that P is orthogonal.

First find the transpose:

$$P^T = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Now compute

$$P^T P = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Therefore,

$$P^T = P^{-1},$$

so P is orthogonal.

3.6 Example 7.3.6: Checking Orthonormal Columns

Given

$$P = \begin{bmatrix} 3/5 & 0 & 4/5 \\ 0 & 1 & 0 \\ -4/5 & 0 & 3/5 \end{bmatrix}.$$

The columns are

$$p_1 = \begin{bmatrix} 3/5 \\ 0 \\ -4/5 \end{bmatrix}, \quad p_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad p_3 = \begin{bmatrix} 4/5 \\ 0 \\ 3/5 \end{bmatrix}.$$

Check lengths:

$$\|p_1\| = \sqrt{(3/5)^2 + 0^2 + (-4/5)^2} = \sqrt{9/25 + 16/25} = 1.$$

$$\|p_2\| = 1.$$

$$\|p_3\| = \sqrt{(4/5)^2 + 0^2 + (3/5)^2} = 1.$$

Check dot products:

$$p_1 \cdot p_2 = 0,$$

$$p_1 \cdot p_3 = \frac{3}{5} \cdot \frac{4}{5} + 0 + \left(-\frac{4}{5}\right) \frac{3}{5} = \frac{12}{25} - \frac{12}{25} = 0,$$

$$p_2 \cdot p_3 = 0.$$

The columns are orthonormal, so P is orthogonal.

3.7 Example 7.3.7: Proving a Matrix Is Orthogonal by Columns

Given

$$P = \begin{bmatrix} 1/3 & -2/\sqrt{5} & -2/(3\sqrt{5}) \\ 2/3 & 1/\sqrt{5} & -4/(3\sqrt{5}) \\ 2/3 & 0 & 5/(3\sqrt{5}) \end{bmatrix}.$$

Let the columns be

$$p_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}, \quad p_2 = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \\ 0 \end{bmatrix}, \quad p_3 = \begin{bmatrix} -2/(3\sqrt{5}) \\ -4/(3\sqrt{5}) \\ 5/(3\sqrt{5}) \end{bmatrix}.$$

Check orthogonality:

$$p_1 \cdot p_2 = \frac{1}{3} \left(-\frac{2}{\sqrt{5}}\right) + \frac{2}{3} \left(\frac{1}{\sqrt{5}}\right) + 0 = 0.$$

$$p_1 \cdot p_3 = \frac{1}{3} \left(-\frac{2}{3\sqrt{5}}\right) + \frac{2}{3} \left(-\frac{4}{3\sqrt{5}}\right) + \frac{2}{3} \left(\frac{5}{3\sqrt{5}}\right) = 0.$$

$$p_2 \cdot p_3 = \left(-\frac{2}{\sqrt{5}}\right) \left(-\frac{2}{3\sqrt{5}}\right) + \left(\frac{1}{\sqrt{5}}\right) \left(-\frac{4}{3\sqrt{5}}\right) + 0 = \frac{4}{15} - \frac{4}{15} = 0.$$

Check lengths:

$$\|p_1\|^2 = \frac{1}{9} + \frac{4}{9} + \frac{4}{9} = 1.$$

$$\|p_2\|^2 = \frac{4}{5} + \frac{1}{5} = 1.$$

$$\|p_3\|^2 = \frac{4}{45} + \frac{16}{45} + \frac{25}{45} = 1.$$

Thus the columns are orthonormal, so P is orthogonal.

3.8 Example 7.3.8: Eigenvectors of a Symmetric Matrix Are Orthogonal

Given

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

Show that eigenvectors corresponding to distinct eigenvalues are orthogonal.

Step 1: Find eigenvalues.

$$\lambda I - A = \begin{bmatrix} \lambda - 3 & -1 \\ -1 & \lambda - 3 \end{bmatrix}.$$

$$\det(\lambda I - A) = (\lambda - 3)^2 - 1 = (\lambda - 4)(\lambda - 2).$$

Thus

$$\lambda = 2, \quad \lambda = 4.$$

For $\lambda = 2$:

$$2I - A = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}.$$

The equation is

$$-x - y = 0,$$

so $x = -y$. Choose

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

For $\lambda = 4$:

$$4I - A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

The equation is

$$x - y = 0,$$

so $x = y$. Choose

$$v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Check the dot product:

$$v_1 \cdot v_2 = 1(1) + (-1)(1) = 0.$$

Therefore, the eigenvectors are orthogonal.

3.9 Example 7.3.9: Which Matrices Are Orthogonally Diagonalizable?

A matrix is orthogonally diagonalizable if and only if it is symmetric.

$$A_1 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

is symmetric, so A_1 is orthogonally diagonalizable.

$$A_2 = \begin{bmatrix} 5 & 2 & -1 \\ 2 & 1 & 8 \\ 1 & 8 & 0 \end{bmatrix}$$

is not symmetric because $a_{13} = -1$ but $a_{31} = 1$. So A_2 is not orthogonally diagonalizable.

$$A_3 = \begin{bmatrix} 3 & 2 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}$$

is not square, so it is not orthogonally diagonalizable.

$$A_4 = \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix}$$

is symmetric, so A_4 is orthogonally diagonalizable.

3.10 Example 7.3.10: Orthogonally Diagonalize a 2×2 Symmetric Matrix

Find an orthogonal matrix P that diagonalizes

$$A = \begin{bmatrix} -2 & 2 \\ 2 & 1 \end{bmatrix}.$$

Step 1: Check symmetry.

$$A^T = A,$$

so A is orthogonally diagonalizable.

Step 2: Find eigenvalues.

$$\lambda I - A = \begin{bmatrix} \lambda + 2 & -2 \\ -2 & \lambda - 1 \end{bmatrix}.$$

$$\begin{aligned} \det(\lambda I - A) &= (\lambda + 2)(\lambda - 1) - 4. \\ &= \lambda^2 + \lambda - 2 - 4 = \lambda^2 + \lambda - 6 = (\lambda + 3)(\lambda - 2). \end{aligned}$$

Thus

$$\lambda = -3, \quad \lambda = 2.$$

For $\lambda = -3$:

$$-3I - A = \begin{bmatrix} -1 & -2 \\ -2 & -4 \end{bmatrix}.$$

The equation is

$$-x - 2y = 0,$$

so $x = -2y$. Choose

$$v_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Normalize:

$$\|v_1\| = \sqrt{(-2)^2 + 1^2} = \sqrt{5}.$$

Thus

$$p_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}.$$

For $\lambda = 2$:

$$2I - A = \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}.$$

The equation is

$$4x - 2y = 0,$$

so $y = 2x$. Choose

$$v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Normalize:

$$\|v_2\| = \sqrt{1^2 + 2^2} = \sqrt{5}.$$

Thus

$$p_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}.$$

Form

$$P = \begin{bmatrix} -2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}.$$

Then

$$P^T A P = \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}.$$

3.11 Example 7.3.11: Orthogonally Diagonalize a 3×3 Symmetric Matrix

Find P that orthogonally diagonalizes

$$A = \begin{bmatrix} 2 & 2 & -2 \\ 2 & -1 & 4 \\ -2 & 4 & -1 \end{bmatrix}.$$

The characteristic polynomial is

$$\det(\lambda I - A) = (\lambda + 6)(\lambda - 3)^2.$$

So

$$\lambda = -6, \quad \lambda = 3.$$

Eigenvector for $\lambda = -6$: Use

$$v_1 = (1, -2, 2).$$

Normalize:

$$\|v_1\| = \sqrt{1^2 + (-2)^2 + 2^2} = 3.$$

Thus

$$u_1 = \left(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}\right).$$

Eigenvectors for $\lambda = 3$: Use two independent eigenvectors

$$v_2 = (2, 1, 0), \quad v_3 = (-2, 0, 1).$$

They are not orthogonal because

$$v_2 \cdot v_3 = 2(-2) + 1(0) + 0(1) = -4.$$

Use Gram-Schmidt.

Let

$$w_2 = v_2 = (2, 1, 0).$$

Then

$$w_3 = v_3 - \left(\frac{v_3 \cdot w_2}{w_2 \cdot w_2}\right) w_2.$$

Compute

$$v_3 \cdot w_2 = -4, \quad w_2 \cdot w_2 = 5.$$

So

$$\begin{aligned} w_3 &= v_3 - \left(-\frac{4}{5}\right) w_2 = v_3 + \frac{4}{5} w_2. \\ w_3 &= (-2, 0, 1) + \left(\frac{8}{5}, \frac{4}{5}, 0\right) = \left(-\frac{2}{5}, \frac{4}{5}, 1\right). \end{aligned}$$

Normalize w_2 :

$$\|w_2\| = \sqrt{5}, \quad u_2 = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}, 0\right).$$

Normalize w_3 :

$$\|w_3\| = \sqrt{\frac{4}{25} + \frac{16}{25} + 1} = \sqrt{\frac{9}{5}} = \frac{3}{\sqrt{5}}.$$

Thus

$$u_3 = \left(-\frac{2}{3\sqrt{5}}, \frac{4}{3\sqrt{5}}, \frac{5}{3\sqrt{5}} \right).$$

Form

$$P = \begin{bmatrix} 1/3 & 2/\sqrt{5} & -2/(3\sqrt{5}) \\ -2/3 & 1/\sqrt{5} & 4/(3\sqrt{5}) \\ 2/3 & 0 & 5/(3\sqrt{5}) \end{bmatrix}.$$

The matching diagonal matrix is

$$D = \begin{bmatrix} -6 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Therefore,

$$P^T A P = D.$$

3.12 Example 7.3.12: Hessian Matrix Note

For a function $f(x, y)$, the Hessian matrix is

$$H = \begin{bmatrix} f_{xx} & f_{yx} \\ f_{xy} & f_{yy} \end{bmatrix}.$$

In many common calculus settings,

$$f_{xy} = f_{yx},$$

so the Hessian is symmetric. This is why symmetric matrices appear in tests for relative maxima and minima.

4 7.4 Applications of Eigenvalues and Eigenvectors

4.1 Example 7.4.1: Population Growth Model

A rabbit population has age classes

$$0 \leq \text{age} < 1, \quad 1 \leq \text{age} < 2, \quad 2 \leq \text{age} \leq 3.$$

The current age distribution is

$$x_1 = \begin{bmatrix} 24 \\ 24 \\ 20 \end{bmatrix}.$$

Rabbits produce offspring at rates 0, 6, 8 in the three age classes. Half survive from the first class to the second, and half survive from the second class to the third.

The Leslie matrix is

$$L = \begin{bmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix}.$$

Use

$$x_2 = Lx_1.$$

Compute:

$$x_2 = \begin{bmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 24 \\ 24 \\ 20 \end{bmatrix}.$$

First row:

$$0(24) + 6(24) + 8(20) = 144 + 160 = 304.$$

Second row:

$$0.5(24) = 12.$$

Third row:

$$0.5(24) = 12.$$

Therefore,

$$x_2 = \begin{bmatrix} 304 \\ 12 \\ 12 \end{bmatrix}.$$

After one year, there are 304 rabbits in class 1, 12 in class 2, and 12 in class 3.

4.2 Example 7.4.2: Stable Age Distribution Vector

For the same Leslie matrix

$$L = \begin{bmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix},$$

find a stable age distribution. The characteristic polynomial is

$$\det(\lambda I - L) = (\lambda + 1)^2(\lambda - 2).$$

Use the positive eigenvalue

$$\lambda = 2.$$

Solve

$$Lx = 2x.$$

Let

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Then

$$Lx = \begin{bmatrix} 6x_2 + 8x_3 \\ 0.5x_1 \\ 0.5x_2 \end{bmatrix}.$$

Set equal to $2x$:

$$6x_2 + 8x_3 = 2x_1,$$

$$0.5x_1 = 2x_2,$$

$$0.5x_2 = 2x_3.$$

From the second equation:

$$x_1 = 4x_2.$$

From the third equation:

$$x_2 = 4x_3.$$

Therefore,

$$x_1 = 4(4x_3) = 16x_3.$$

Let $x_3 = t$. Then

$$x = \begin{bmatrix} 16t \\ 4t \\ t \end{bmatrix} = t \begin{bmatrix} 16 \\ 4 \\ 1 \end{bmatrix}.$$

So the stable age distribution ratio is

$$16 : 4 : 1.$$

Check with $t = 2$:

$$x = \begin{bmatrix} 32 \\ 8 \\ 2 \end{bmatrix}.$$

$$Lx = \begin{bmatrix} 0 & 6 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 32 \\ 8 \\ 2 \end{bmatrix} = \begin{bmatrix} 64 \\ 16 \\ 4 \end{bmatrix} = 2 \begin{bmatrix} 32 \\ 8 \\ 2 \end{bmatrix}.$$

This confirms $Lx = 2x$.

4.3 Example 7.4.3: Solving a Diagonal Differential System

Solve

$$y'_1 = 4y_1, \quad y'_2 = -y_2, \quad y'_3 = 2y_3.$$

Use the basic rule:

$$y' = ky \implies y = Ce^{kt}.$$

Therefore,

$$y_1 = C_1 e^{4t},$$

$$y_2 = C_2 e^{-t},$$

$$y_3 = C_3 e^{2t}.$$

4.4 Example 7.4.4: Solving a Non-Diagonal Differential System

Solve

$$y_1' = 3y_1 + 2y_2, \quad y_2' = 6y_1 - y_2.$$

Write

$$y' = Ay, \quad A = \begin{bmatrix} 3 & 2 \\ 6 & -1 \end{bmatrix}.$$

Step 1: Find eigenvalues.

$$\begin{aligned} \lambda I - A &= \begin{bmatrix} \lambda - 3 & -2 \\ -6 & \lambda + 1 \end{bmatrix}. \\ \det(\lambda I - A) &= (\lambda - 3)(\lambda + 1) - 12. \\ &= \lambda^2 - 2\lambda - 3 - 12 = \lambda^2 - 2\lambda - 15. \end{aligned}$$

Factor:

$$\lambda^2 - 2\lambda - 15 = (\lambda - 5)(\lambda + 3).$$

So

$$\lambda = -3, \quad \lambda = 5.$$

For $\lambda = -3$:

$$-3I - A = \begin{bmatrix} -6 & -2 \\ -6 & -2 \end{bmatrix}.$$

Equation:

$$-6x_1 - 2x_2 = 0.$$

Divide by -2 :

$$3x_1 + x_2 = 0,$$

so

$$x_2 = -3x_1.$$

Choose

$$p_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}.$$

For $\lambda = 5$:

$$5I - A = \begin{bmatrix} 2 & -2 \\ -6 & 6 \end{bmatrix}.$$

Equation:

$$2x_1 - 2x_2 = 0,$$

so

$$x_1 = x_2.$$

Choose

$$p_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Form

$$P = \begin{bmatrix} 1 & 1 \\ -3 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} -3 & 0 \\ 0 & 5 \end{bmatrix}.$$

Use $y = Pw$. Then

$$w' = Dw.$$

So

$$w_1' = -3w_1, \quad w_2' = 5w_2.$$

Solve:

$$w_1 = C_1 e^{-3t}, \quad w_2 = C_2 e^{5t}.$$

Convert back:

$$y = Pw = \begin{bmatrix} 1 & 1 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}.$$

Thus

$$\begin{aligned} y_1 &= w_1 + w_2 = C_1 e^{-3t} + C_2 e^{5t}, \\ y_2 &= -3w_1 + w_2 = -3C_1 e^{-3t} + C_2 e^{5t}. \end{aligned}$$

4.5 Example 7.4.5: Matrix of a Quadratic Form

Find the matrix of each quadratic form.

Part A:

$$4x^2 + 9y^2 - 36 = 0.$$

Here

$$a = 4, \quad b = 0, \quad c = 9.$$

The quadratic form matrix is

$$A = \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}.$$

Part B:

$$13x^2 - 10xy + 13y^2 - 72 = 0.$$

Here

$$a = 13, \quad b = -10, \quad c = 13.$$

So

$$A = \begin{bmatrix} 13 & -10/2 \\ -10/2 & 13 \end{bmatrix} = \begin{bmatrix} 13 & -5 \\ -5 & 13 \end{bmatrix}.$$

4.6 Example 7.4.6: Rotation of a Conic

Eliminate the xy -term in

$$13x^2 - 10xy + 13y^2 - 72 = 0.$$

Step 1: Form the quadratic form matrix.

$$A = \begin{bmatrix} 13 & -5 \\ -5 & 13 \end{bmatrix}.$$

Step 2: Find eigenvalues.

$$\lambda I - A = \begin{bmatrix} \lambda - 13 & 5 \\ 5 & \lambda - 13 \end{bmatrix}.$$

$$\det(\lambda I - A) = (\lambda - 13)^2 - 25 = (\lambda - 18)(\lambda - 8).$$

Thus

$$\lambda = 8, \quad \lambda = 18.$$

For $\lambda = 8$:

$$8I - A = \begin{bmatrix} -5 & 5 \\ 5 & -5 \end{bmatrix}.$$

Equation:

$$-5x + 5y = 0,$$

so $x = y$. Choose $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and normalize:

$$u_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}.$$

For $\lambda = 18$:

$$18I - A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}.$$

Equation:

$$x + y = 0,$$

so choose $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and normalize:

$$u_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}.$$

Form

$$P = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

This corresponds to a 45° rotation.

The rotated equation becomes

$$8(x')^2 + 18(y')^2 - 72 = 0.$$

Divide by 72:

$$\frac{(x')^2}{9} + \frac{(y')^2}{4} = 1.$$

This is an ellipse.

4.7 Example 7.4.7: Rotation of a Conic with Linear Terms

Eliminate the xy -term:

$$3x^2 - 10xy + 3y^2 + 16\sqrt{2}x - 32 = 0.$$

Step 1: Matrix of the quadratic form.

$$A = \begin{bmatrix} 3 & -5 \\ -5 & 3 \end{bmatrix}.$$

Step 2: Eigenvalues.

$$\lambda I - A = \begin{bmatrix} \lambda - 3 & 5 \\ 5 & \lambda - 3 \end{bmatrix}.$$

$$\det(\lambda I - A) = (\lambda - 3)^2 - 25 = (\lambda - 8)(\lambda + 2).$$

So

$$\lambda = 8, \quad \lambda = -2.$$

Use normalized eigenvectors

$$u_1 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right), \quad u_2 = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right).$$

Thus

$$P = \begin{bmatrix} -1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}.$$

This corresponds to a 135° rotation.

Step 3: Transform the linear part. The linear row is

$$[d \ e] = [16\sqrt{2} \ 0].$$

Then

$$[d \ e]P = [16\sqrt{2} \ 0] \begin{bmatrix} -1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} = [-16 \ -16].$$

So the rotated equation is

$$8(x')^2 - 2(y')^2 - 16x' - 16y' - 32 = 0.$$

Step 4: Complete the square. Group x' terms:

$$8(x')^2 - 16x' = 8[(x')^2 - 2x'] = 8[(x' - 1)^2 - 1].$$

So

$$8(x')^2 - 16x' = 8(x' - 1)^2 - 8.$$

Group y' terms:

$$-2(y')^2 - 16y' = -2[(y')^2 + 8y'].$$

Complete the square:

$$(y')^2 + 8y' = (y' + 4)^2 - 16.$$

So

$$-2(y')^2 - 16y' = -2(y' + 4)^2 + 32.$$

Substitute:

$$8(x' - 1)^2 - 8 - 2(y' + 4)^2 + 32 - 32 = 0.$$

Simplify:

$$8(x' - 1)^2 - 2(y' + 4)^2 - 8 = 0.$$

Move -8 :

$$8(x' - 1)^2 - 2(y' + 4)^2 = 8.$$

Divide by 8:

$$(x' - 1)^2 - \frac{(y' + 4)^2}{4} = 1.$$

This is a hyperbola.

4.8 Example 7.4.8: Rotation of a Quadric Surface

Eliminate the xz -term:

$$5x^2 + 4y^2 + 5z^2 + 8xz - 36 = 0.$$

Step 1: Form the quadratic form matrix. The xz coefficient is 8, so the matrix entries are $8/2 = 4$ in the xz and zx positions:

$$A = \begin{bmatrix} 5 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 5 \end{bmatrix}.$$

Step 2: Use eigenvalues. The eigenvalues are

$$\lambda = 1, \quad 4, \quad 9.$$

After rotation, the equation becomes

$$(x')^2 + 4(y')^2 + 9(z')^2 - 36 = 0.$$

Step 3: Standard form.

$$(x')^2 + 4(y')^2 + 9(z')^2 = 36.$$

Divide by 36:

$$\frac{(x')^2}{36} + \frac{(y')^2}{9} + \frac{(z')^2}{4} = 1.$$

This is an ellipsoid.

4.9 Example 7.4.9: Maximum and Minimum with a Diagonal Matrix

Find the maximum and minimum of

$$f(x_1, x_2) = 9x_1^2 + 5x_2^2$$

subject to

$$x_1^2 + x_2^2 = 1.$$

The matrix is

$$A = \begin{bmatrix} 9 & 0 \\ 0 & 5 \end{bmatrix}.$$

The eigenvalues are

$$\lambda = 9, \quad \lambda = 5.$$

For a quadratic form $x^T Ax$ subject to $\|x\|^2 = 1$, the maximum is the greatest eigenvalue and the minimum is the least eigenvalue.

Therefore,

$$\text{maximum} = 9, \quad \text{minimum} = 5.$$

The maximum occurs at

$$(1, 0) \quad \text{or} \quad (-1, 0),$$

and the minimum occurs at

$$(0, 1) \quad \text{or} \quad (0, -1).$$

4.10 Example 7.4.10: Maximum and Minimum with a Cross Term

Find the maximum and minimum of

$$z = 7x_1^2 + 6x_1x_2 + 7x_2^2$$

subject to

$$x_1^2 + x_2^2 = 1.$$

Step 1: Matrix of the quadratic form. The cross-term coefficient is 6, so split it into 3 and 3:

$$A = \begin{bmatrix} 7 & 3 \\ 3 & 7 \end{bmatrix}.$$

Step 2: Find eigenvalues.

$$\lambda I - A = \begin{bmatrix} \lambda - 7 & -3 \\ -3 & \lambda - 7 \end{bmatrix}.$$

$$\det(\lambda I - A) = (\lambda - 7)^2 - 9 = (\lambda - 10)(\lambda - 4).$$

So

$$\lambda = 10, \quad \lambda = 4.$$

Therefore,

$$\text{maximum} = 10, \quad \text{minimum} = 4.$$

Maximum point: For $\lambda = 10$:

$$10I - A = \begin{bmatrix} 3 & -3 \\ -3 & 3 \end{bmatrix}.$$

Equation:

$$x_1 = x_2.$$

A normalized eigenvector is

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right).$$

So the maximum occurs at

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

and also at its negative.

Minimum point: For $\lambda = 4$:

$$4I - A = \begin{bmatrix} -3 & -3 \\ -3 & -3 \end{bmatrix}.$$

Equation:

$$x_1 + x_2 = 0.$$

A normalized eigenvector is

$$\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right).$$

So the minimum occurs at

$$\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

and also at its negative.

4.11 Example 7.4.11: Constrained Optimization with Change of Variables

Find the maximum and minimum of

$$z = 4xy$$

subject to

$$9x^2 + 4y^2 = 36.$$

The constraint is not $x^2 + y^2 = 1$, so change variables:

$$x = 2x', \quad y = 3y'.$$

Substitute into the constraint:

$$9(2x')^2 + 4(3y')^2 = 36.$$

$$36(x')^2 + 36(y')^2 = 36.$$

Divide by 36:

$$(x')^2 + (y')^2 = 1.$$

Rewrite the objective:

$$z = 4xy = 4(2x')(3y') = 24x'y'.$$

The matrix of the quadratic form is

$$A = \begin{bmatrix} 0 & 12 \\ 12 & 0 \end{bmatrix}.$$

Find eigenvalues:

$$\lambda I - A = \begin{bmatrix} \lambda & -12 \\ -12 & \lambda \end{bmatrix}.$$

$$\det(\lambda I - A) = \lambda^2 - 144 = (\lambda - 12)(\lambda + 12).$$

So

$$\lambda = 12, \quad \lambda = -12.$$

Therefore,

$$\text{maximum} = 12, \quad \text{minimum} = -12.$$

Maximum point: For $\lambda = 12$, the eigenvector satisfies

$$x' = y'.$$

Use the normalized vector

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right).$$

Convert back:

$$x = 2x' = 2 \left(\frac{1}{\sqrt{2}} \right) = \sqrt{2},$$

$$y = 3y' = 3 \left(\frac{1}{\sqrt{2}} \right) = \frac{3}{\sqrt{2}}.$$

So the maximum occurs at

$$\left(\sqrt{2}, \frac{3}{\sqrt{2}} \right).$$

Minimum point: For $\lambda = -12$, the eigenvector satisfies

$$x' = -y'.$$

Use the normalized vector

$$\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right).$$

Convert back:

$$x = 2 \left(\frac{1}{\sqrt{2}} \right) = \sqrt{2},$$

$$y = 3 \left(-\frac{1}{\sqrt{2}} \right) = -\frac{3}{\sqrt{2}}.$$

So the minimum occurs at

$$\left(\sqrt{2}, -\frac{3}{\sqrt{2}} \right).$$

Final Chapter 7 Checklist

Eigenvalues and eigenvectors:

$$Ax = \lambda x, \quad \det(\lambda I - A) = 0, \quad (\lambda I - A)x = 0.$$

Diagonalization:

A is diagonalizable $\iff A$ has n linearly independent eigenvectors.

$$P^{-1}AP = D.$$

Symmetric matrices:

$$A = A^T.$$

Symmetric matrices are orthogonally diagonalizable.

Orthogonal matrices:

$$P^{-1} = P^T.$$

Orthogonal diagonalization:

$$P^TAP = D.$$

Population growth:

$$x_{j+1} = Lx_j.$$

Stable age distribution:

$$Lx = \lambda x.$$

Differential systems:

$$y' = Ay, \quad y = Pw, \quad w' = Dw.$$

Quadratic forms in two variables:

$$A = \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix}.$$

Quadratic forms in three variables:

$$A = \begin{bmatrix} a & d/2 & e/2 \\ d/2 & b & f/2 \\ e/2 & f/2 & c \end{bmatrix}.$$

Constrained optimization: For

$$f(x) = x^T Ax, \quad \|x\|^2 = 1, \\ \text{maximum} = \lambda_{\max}, \quad \text{minimum} = \lambda_{\min}.$$